

Manish Kumar Soni

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SUMMARY

Final-year B.Tech student and self-directed full-stack engineer. I build systems end to end and care about what fails quietly: contracts shared between client and server, correct HTTP semantics, untrusted code that cannot reach a secret, and money arithmetic that does not lose a cent to a float. Seeking a **Software Engineer (New Grad, 2027)** role in full-stack or AI product engineering.

EDUCATION

VIT Bhopal University

2023 – 2027 (Expected)

B.Tech, Computer Science and Engineering

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES2023), Python, SQL, HTML5/CSS3, GLSL, Bash

Frontend: React 19, Next.js 16 (App Router, Server Actions), Tailwind CSS v4, shadcn/ui, Zustand, TanStack Query, React Hook Form, Zod, Three.js, WebGL, Vite

Backend: Node.js, Express 5, FastAPI, REST API design, Prisma 7, JWT access/refresh auth, OAuth 2.0 (Passport), Stripe, OpenAI / Groq / OpenRouter SDKs

Data & Infra: PostgreSQL, Redis, SQLite/libSQL (Turso), schema design & migrations, Docker Compose, GitHub Actions CI, Turborepo, pnpm workspaces, Vercel, Git

Practices: integration testing (Vitest, Supertest, Pytest, Testing Library), typed error handling, provider abstraction with graceful fallback, process isolation for untrusted code

PROJECTS

HireFlow — Recruitment & Applicant Tracking Platform

github.com/Mu0810/hireflow-ai

TypeScript, Next.js 16, React 19, Express 5, PostgreSQL, Prisma 7, Docker, Turborepo

- Architected a **Turborepo + pnpm monorepo** — Next.js 16 client, Express 5 REST API, shared package — where Zod schemas and domain types are declared once for both runtimes, so a payload change surfaces as a type error rather than a runtime 400.
- Modelled the hiring lifecycle in a **25-model PostgreSQL schema** (12 enums, 9 migrations) driving ~1,800 lines of service logic across 11 REST route modules: companies, jobs, applications, coding tests, interviews, analytics, and subscriptions.
- Closed a **secret-disclosure hole** in the coding-test grader: candidate code ran in-process under Node's vm, where a one-line escape reads the database URL and both JWT signing secrets. Moved execution to an environment-stripped, heap-capped child process killed on a wall-clock timeout, then **proved containment by running the escape** and asserting the recovered environment was empty.
- Replaced blanket 400 responses with **typed errors** mapped to 401/403/404/409 across all 66 service failure paths, and diagnosed two faults that had prevented **CI from ever running a test** — now green on Supertest suites against a live Postgres container.

BuyWise AI — AI Shopping Intelligence Platform

github.com/Mu0810/buywise-ai

Next.js 16, React 19, TypeScript (strict), Prisma 7, PostgreSQL, Redis, Stripe, OpenAI

- Built an AI assistant turning plain-language queries into value-scored, cross-retailer recommendations, asking clarifying follow-ups before recommending rather than listing raw results.
- Designed a **provider abstraction** so new retailers plug in without touching business logic, mirrored for AI — an OpenAI engine beside a **rule-based fallback** that keeps the product fully usable with no API key configured.
- Shipped Stripe subscriptions with webhooks and a billing portal, Supabase Auth, and Redis caching across 13 feature modules, with Zod validation at every boundary.

Healthvibe — Energy-Availability & Nutrition Tracker

github.com/Mu0810/health-webapp

Next.js 16, TypeScript, Prisma, libSQL/Turso, OpenRouter

- Tracks **Energy Availability** — intake minus exercise burn, over fat-free mass — rather than calories alone, driving the interface from its green/amber/red thresholds.
- Implemented BMI, **Mifflin-St Jeor** BMR, TDEE, and macro targets as pure functions with no React and no I/O, so the health maths is unit-tested directly instead of through the UI.
- Added photo-based food logging: an image-capable model returns macros with a confidence rating, behind an **ordered model fallback chain** that survives the free tier's unpredictable rate limiting instead of failing the request.

Online Chess — Zero-Dependency Chess Engine & AI Opponent

github.com/Mu0810/chess

JavaScript, HTML5, Web Audio API, Node.js

- Wrote a **complete rules engine from scratch** (693 lines): legal move generation, castling, en passant, promotion, check/checkmate/stalemate, plus threefold-repetition, fifty-move, and insufficient-material draws, with PGN/FEN import-export.
- Implemented the opponent as a **negamax search with alpha-beta pruning, MVV-LVA move ordering, and quiescence extension** across four levels that vary search depth, aspiration window, and evaluation noise — with zero runtime dependencies and no build step.

ADDITIONAL PROJECTS

Meridian Reserve | [Source](#) | Hotel booking flow with money maths in integer cents, because $850 * 1.15$ rounds down and silently undercharges a dollar, and dates anchored to noon UTC so timezone offsets cannot corrupt night counts. **17 test suites**.

Real-time graphics | [Super Plumber 3D](#) | [WebGL Aurora](#) | A Three.js platformer with no art assets or physics engine — geometry, collision and audio generated in code — plus a hand-written **GLSL** noise shader and a 240-particle field reduced from 57,600 pairwise checks per frame to **O(n)** by spatial hashing.

AI Research Agent | [Live](#) | **FastAPI + Groq** inference with web-search, scraping and PDF-parsing tools, over PostgreSQL.